

California Housing Dataset – Short ML Project Report

This report summarizes the exploratory data analysis (EDA), machine learning model, evaluation metrics, and possible improvements for the California Housing Dataset project. A Linear Regression model was used to predict housing prices based on multiple socioeconomic features.

1. Exploratory Data Analysis (EDA)

The dataset contains features such as median income, house age, average rooms, average occupancy, latitude, and longitude. The dataset was checked for:

- Missing values
- Statistical summary using describe()
- Correlation between variables
- Feature distributions using histograms

Key Findings:

- No missing values were found.
- Median income showed the strongest positive correlation with house prices.
- Geographic features such as latitude and longitude influenced pricing patterns.
- Some variables contained outliers and skewed distributions.

2. Model Development

A Linear Regression model from scikit-learn was trained using an 80:20 train-test split. The target variable was the house price, while all remaining columns were used as input features.

Steps performed:

- Dataset split into training and testing sets
- Model trained using LinearRegression()
- Predictions generated on the test dataset

3. Evaluation Metrics

Metric	Purpose
MAE	Measures average prediction error
RMSE	Penalizes larger prediction errors
R ² Score	Indicates model accuracy and fit

The Linear Regression model achieved moderate prediction performance. RMSE and MAE values indicated that prediction errors were present, while the R² score showed that the model explained a reasonable portion of the variance in house prices.

4. Improvement Ideas

The following techniques can improve model performance:

- Use advanced models such as Random Forest, XGBoost, or Gradient Boosting.
- Perform feature engineering and remove outliers.

- Apply feature scaling and hyperparameter tuning.
- Use cross-validation for more reliable evaluation.
- Add visualization plots such as residual plots and prediction scatter plots.

Conclusion:

Linear Regression provides a good baseline model for predicting California housing prices. Median income was identified as the most influential feature. While the model performs reasonably well, more advanced machine learning algorithms can further improve prediction accuracy.